

```
$ emulator @myandy_2.2 -shell
# sqlite3
sqlite3> .help
```

Generally, when you have a database-based Android application, the application database is stored in `/data/data/<application package>/databases`. For example, the email application's database is in the folder `/data/data/com.android.email/databases`.

At the `sqlite3` prompt, you can perform data definition and manipulation statements with SQL syntax. For example:

```
sqlite3> create table employee(
emp_id int,
emp_name varchar(20)
);
sqlite3> .tables
employee
```

Traceview

This tool is used to profile applications. To be able to use it, there are some requirements for your application—its code must have a `Debug.startTracingMethod(String tracename)` method call at the beginning, and a `Debug.stopTracingMethod()` call at the end of the method. This method stores the tracing information onto an SD card—hence, your emulator instance should have an SD card image attached. The tracing writes a file at `/sdcard/<tracename>.trace` and hence we need to give the application permission to write the file on the SD card. This permission is given in `AndroidManifest`, as:

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
```

When the application is run in the

emulator instance, you need to retrieve the trace file and feed it to the `traceview` utility, as follows:

```
$ adb pull /sdcard/HelloWorld.trace /tmp
$ traceview /tmp/HelloWorld.trace
```

Now, you will be presented with a `traceview` window as shown in Figure 11, which is divided into two sections:

- The timeline section, which shows you the execution time of each thread in corresponding rows, and
- The profile section, which shows you the time that is spent by the CPU inside a method, with some statistics included in separate columns.

Some other tools that are used internally are:

- `dx`: To convert a `.class` file to a `.dex` file.
- `Monkey`: To stress-test an Android application, this device console tool issues random events to a given application, such as touch, gesture, or click events.

To run `Monkey` on the default Android email application, in a new emulator instance, issue the following command:

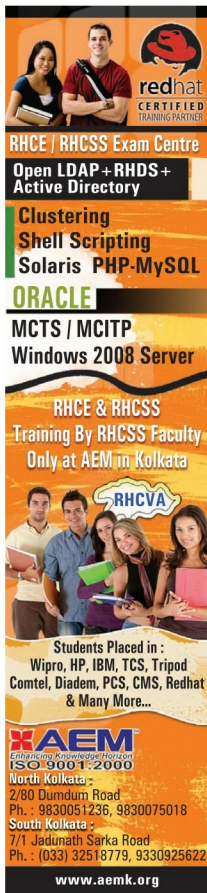
```
$ emulator @andy_2.2
$ adb shell monkey -p com.android.email
-v 500
```

Here, `Monkey` is passed on the application package name (`com.android.email`), and the number of tests (500) to run.

I hope you enjoyed this article, and that the knowledge I've shared is helpful in your quest to learn more about Android. I will be back with other interesting information on Android Application Development in upcoming issues. We will also discuss building advanced applications on Android and for Android. 

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